

MODULE 9.1

Defending Systems and Devices

Endpoint Security | Cisco Networking Academy

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Cybersecurity Essentials

Topics Covered:

OS Security • Antimalware Programs • Patch Management

Endpoint Security • Host Encryption • Boot Integrity

Apple Security Features • Managing Device Threats • Physical Protection

9.1.1 – 9.1.2 Operating System Security

The OS is the first line of defense in network security

9.1.1

Why OS Security Matters

The operating system plays a critical role in maintaining network security and is the target of many attacks. Securing the OS is the first responsibility of cybersecurity professionals.

REMOVE

Unnecessary
Programs & Services

PATCH

Install Security
Updates Promptly

CONFIGURE

Set Secure OS
Settings & Policies

9.1.1 – 9.1.2 Operating System Security

The OS is the first line of defense in network security

9.1.2

A Good Administrator Will:

- Configure the OS to protect against outside threats
- Remove unnecessary programs and services
- Install security patches and updates in a timely manner to correct faults and mitigate risks

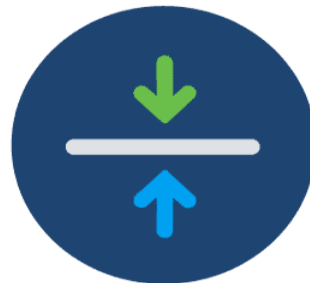
A systematic approach:

It's important to have a systematic approach in place for addressing system updates. An organization should:

- establish procedures for monitoring security-related information
- evaluate updates for applicability
- plan the installation of application updates and patches
- install updates using a documented plan.

A baseline:

Another critical way to secure an operating system is to identify potential vulnerabilities. To do this, establish a baseline to compare how a system is performing against baseline expectations.



9.1.3 Types of Antimalware Programs

Malware = viruses, worms, Trojans, keyloggers, spyware, adware

It's important to protect devices using reputable antimalware software. Different tools target different threats:

	Adware Protection	Looks for programs that display unwanted advertising in popup boxes
	Trusted Advisor	Warns the user about unsafe programs or websites
	Phishing Protection	Blocks known phishing IPs and warns about suspicious emails
	Antivirus	Monitors for viruses, warns when detected, quarantines or deletes it
	Anti-spyware	Scans for keyloggers and other spyware that record keystrokes to steal passwords

9.1.4 Points to Remember: Antimalware

Key warnings and best practices for antimalware protection

Watch Out for Rogue Antivirus

Malicious rogue AV products appear as fake Windows warnings while browsing. They claim malware is present and prompt the user to 'clean' it — but clicking anywhere may download real malware instead.



Fileless Attacks Are Hard to Detect

Fileless malware uses legitimate programs (e.g. Windows PowerShell) and loads directly into memory, leaving no footprint. It ends when the system reboots — hard to detect with traditional AV.

Scripts Can Be Malware

Scripting languages like Python, Bash, or VBA (Microsoft macros) can be weaponised to create malicious scripts. These may bypass signature-based detection.

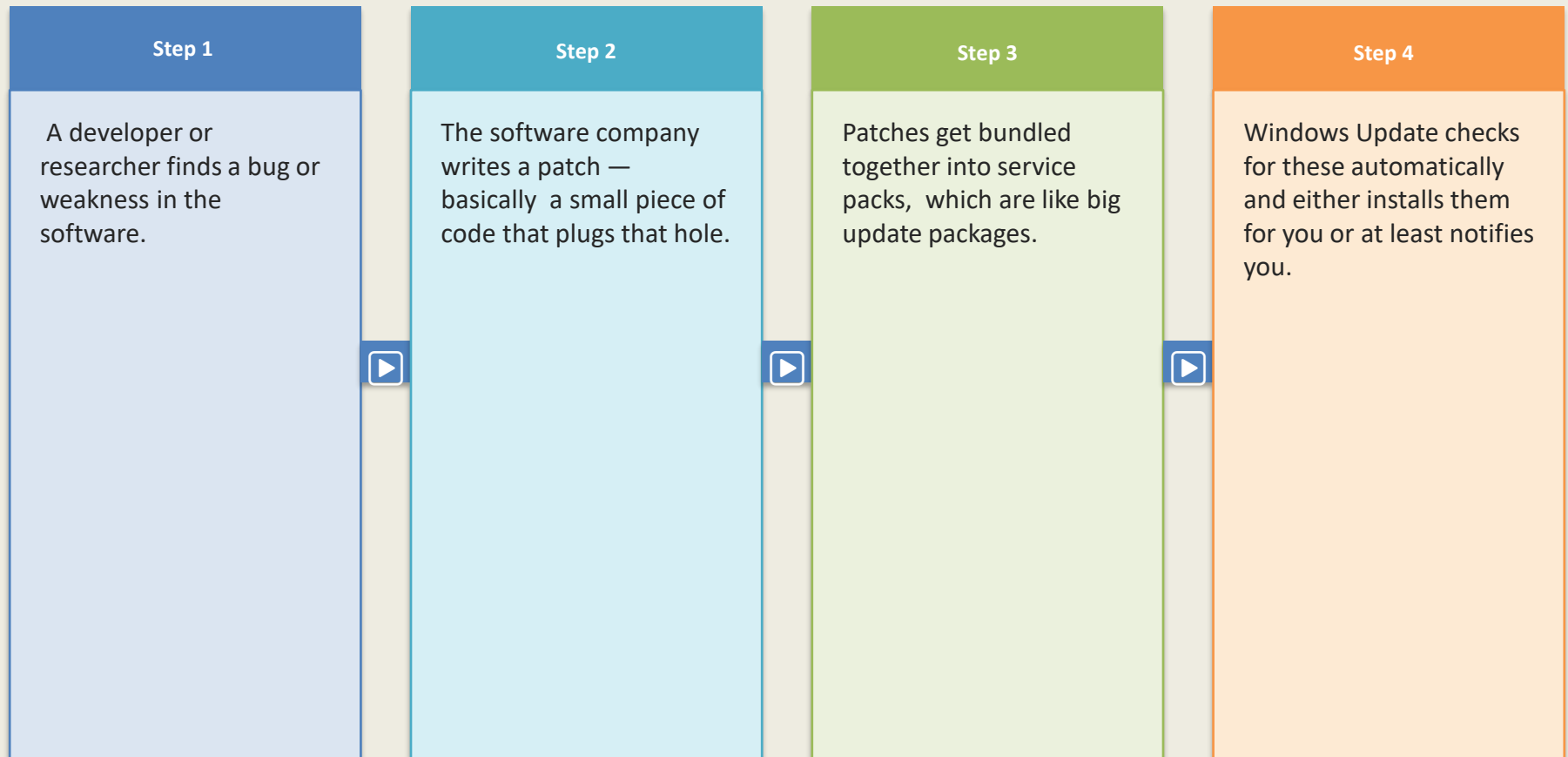
Always Remove Unapproved Software

Unapproved or non-compliant software may be accidentally or deliberately installed. Even if not malicious, it can violate security policy and interfere with the organisation's network. Remove immediately.

9.1.5 Patch Management

Keep systems secure by regularly installing patches and updates

Cybercriminals relentlessly exploit weaknesses. Staying ahead requires regular patching — many attacks could have been avoided with the latest service packs installed.



9.1.6 Endpoint Security

Host-based solutions protect individual devices from attacks

A host-based solution is a software application that runs on a local device (endpoint) to protect it, working with the OS to help prevent attacks.

Host-based Firewall

Restricts incoming/outgoing network activity. Filters data packets. Windows Firewall is built-in. Manages inbound/outbound rules via ports.

HIDS — (Host Intrusion — Detection)

Monitors system calls, file access, and registry for malicious activity. Stores logs locally. Resource-intensive. Cannot monitor network traffic not reaching the host.

HIPS — (Host Intrusion — Prevention)

Monitors for known attacks and anomalies in bandwidth, protocols, ports. Can send alarms, log activity, reset connections, or drop malicious packets.

EDR — (Endpoint Detection — & Response)

Continuously monitors and collects endpoint data, analyzes it, and responds to threats. Goes beyond antivirus — can actively find threats on the device.

DLP — (Data Loss — Prevention)

Provides centralized control to ensure sensitive data is not lost, misused, or accessed by unauthorized users.

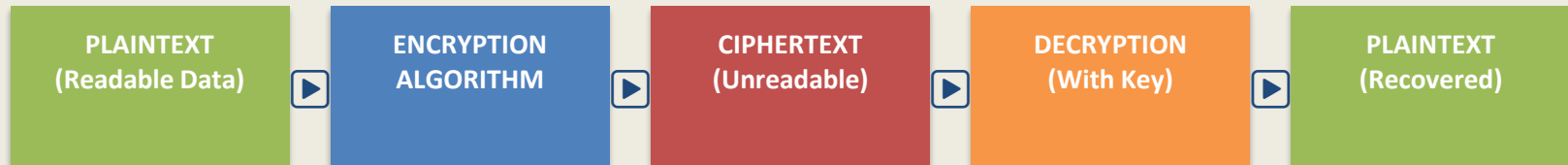
NGFW — (Next-Generation — Firewall)

Combines traditional firewall with advanced filtering functions, e.g. application firewall with deep packet inspection (DPI) and intrusion protection (IPS).

9.1.7 – 9.1.8 Encryption & Host Encryption

Protecting data at rest using encryption technologies

Encryption uses a complex algorithm to transform readable data (plaintext) into unreadable data (ciphertext). A special key reverses this process.



Windows EFS (Encrypting File System)

- Encrypt individual files, folders, or entire hard drive
- Built into Windows OS
- File-level or folder-level encryption

BitLocker (Full Disk Encryption)

- Encrypts entire drive including temp files & memory
- Requires Trusted Platform Module (TPM) in BIOS
- TPM stores encryption keys, certificates, passwords
- BitLocker To Go: encrypts removable drives (no TPM)

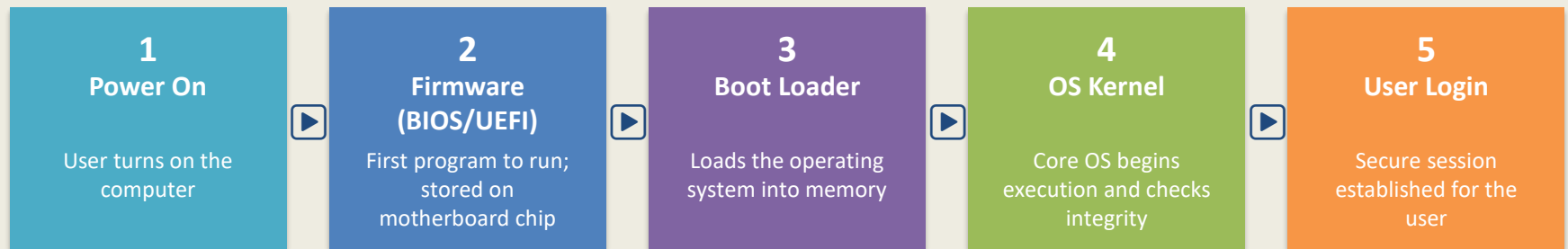
Self-Encrypting Drive (SED)

- Automatically encrypts all data written to drive
- Hardware-based — no OS intervention required
- Prevents attackers from accessing data via their OS

9.1.9 Boot Integrity

Ensuring systems remain secure even during startup

Attackers can strike during the short window when a system starts up. Boot integrity ensures the system can be trusted and has not been altered while loading.



BIOS vs UEFI

BIOS (Basic Input/Output System)

- First program run at power-on
- Stored on small chip on motherboard
- 32-bit mode only
- Older standard — less secure
- Limited boot verification

UEFI (Unified Extensible Firmware Interface)

- Newer, modern replacement for BIOS
- Defines standard interface: OS ↔ firmware ↔ devices
- Runs in 64-bit mode — preferred over BIOS
- Supports Secure Boot (verifies OS signatures)
- Better hardware compatibility

9.1.10 Apple System Security Features

macOS hardware and software security capabilities

Apple provides hardware and macOS security features that offer robust endpoint protection, complementing Windows and Linux security.

Security Hardware / (Secure Enclave)

Special CPU with dedicated AES encryption engine on a System-on-Chip.

Encrypted Storage / (FileVault / Data Protection)

Hardware AES engine encrypts/decrypts files without exposing keys to OS or apps.

Secure Boot / (Boot ROM)

Protects low-level hardware; only genuine, unaltered Apple OS software may run.

Secure Biometrics / (Touch ID / Face ID)

Biometric data processed in security hardware — segregated from OS and running apps.

Find My Mac

Location tracking for lost/stolen devices with remote lock and storage erase.

XProtect

Signature-based malware detection; alerts user and offers removal option.

MRT / (Malware Removal Tool)

Detects/removes existing infections using auto-updated rules. Runs at startup and login.

Gatekeeper

Only authentic, digitally-signed, Apple-notarized software may be installed.

9.1.11 Managing Device Threats

Identifying threats and applying appropriate countermeasures

For each device domain threat, there is a corresponding countermeasure to manage and reduce risk:

THREAT	COUNTERMEASURE
Unpatched Software	Patch Management — install updates and security patches promptly
User Downloads	User Awareness Training and Download Policies / Whitelisting
Malware	Antimalware / Antivirus Software with real-time protection
Unattended Devices	Screen Lock Policies and Physical Security Controls
AUP Violation	Acceptable Use Policy enforcement and employee monitoring
Unauthorized Media	Disable USB ports / Implement removable media policies

9.1.12 Physical Protection of Devices

Security measures to protect hardware and telecommunications rooms

Software defenses are not enough — physical access can bypass all digital controls. Physical security is the last line of defence for hardware and sensitive rooms.

Cable Locks

- Secure devices to desks or fixed objects
- Prevents theft of laptops and workstations
- Kensington lock standard widely used

Cipher Locks / Access Control

- Lock telecommunications rooms
- Combination or PIN-based locks
- Restrict access to authorized personnel only

Faraday Cages / Security Cages

- Metal enclosures block electromagnetic fields
- Prevent wireless eavesdropping or attacks
- Used around sensitive equipment

RFID & Surveillance

- RFID tags track physical assets in real time
- CCTV cameras monitor access points
- Logging and auditing of physical access events

Remember: A strong security posture requires BOTH software controls AND physical security measures working together.

Module 9.1 — Summary

OS Security	Harden OS: remove unused programs/services, apply patches promptly.	Antimalware	5 types: adware protection, trusted advisor, phishing, antivirus, anti-spyware.
Patch Management	Patches fix vulnerabilities; service packs bundle updates; use Windows Update.	Endpoint Security	6 host-based solutions: Firewall, HIDS, HIPS, EDR, DLP, NGFW.
Encryption	EFS (files), BitLocker (FDE + TPM), BitLocker To Go, Self-Encrypting Drive.	Boot Integrity	BIOS vs UEFI — Secure Boot ensures OS authenticity during startup.
Apple Security	Secure Enclave, FileVault, Secure Boot, XProtect, MRT, Gatekeeper.	Physical Security	Cable locks, cipher locks, Faraday cages, RFID tracking, CCTV.